IILM UNIVERSITY
GREATER NOIDA, UTTAR PRADESH

AN INTERNSHIP REPORT

ON

CLAIM PRO — INSURANCE CLAIMS MANAGEMENT SYSTEM

A High-Performance Responsive Single-Page Application (SPA)
Built with HTML5, CSS3, Vanilla JavaScript (ES6+), and Browser LocalStorage

Submitted in partial fulfillment of the requirements
for the award of the degree of

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING

School of Computer Science and Engineering
IILM University, Greater Noida, Uttar Pradesh

SUBMITTED BY:
Shreshth Chauhan
Roll No.: 2410030045
Section: 3CSE14
Batch: 2024–2028
B.Tech (CSE)

INTERNSHIP DETAILS:
Organization: Data Alcott Systems
Student Code: DAS012321
Task ID: #5110 | Internship ID: 5696
Duration: 21–28 Oct 2026
Portal: FreeInternships.in

1. CANDIDATE'S DECLARATION

I, SHRESHTH CHAUHAN (Roll No.: 2410030045, Section: 3CSE14), student of Bachelor of Technology in Computer Science and Engineering at School of Computer Science and Engineering, IILM University, Greater Noida, hereby declare that the internship project report entitled "CLAIM PRO — Insurance Claims Management System" has been completed by me to fulfill the academic curriculum requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering.

I affirm that this report represents my own original technical work undertaken during the Web Development Internship (Task ID: #5110, Internship ID: 5696) hosted by Data Alcott Systems from 21 October 2026 to 28 October 2026. All the technical literature, code references, web specifications, and resources have been appropriately cited. This report has not been submitted previously to any other university or institution for any academic degree, diploma, or certificate award.

Signature of the Candidate: _____

Student Name: Shreshth Chauhan
University Roll Number: 2410030045
Class Section: 3CSE14
Degree Program: Bachelor of Technology (Computer Science & Engineering)
Department: School of Computer Science and Engineering
Institution: IILM University, Greater Noida, Uttar Pradesh
Date: October 2026

2. ACKNOWLEDGEMENT

I take this opportunity to express my heartfelt gratitude to everyone who supported, guided, and encouraged me throughout my Web Development Internship. The practical exposure gained during this period has been invaluable in expanding my technical expertise, problem-solving abilities, and engineering discipline.

I extend my sincere appreciation to the technical team and management at Data Alcott Systems and FreeInternships.in for granting me the opportunity to work on the CLAIM PRO Insurance Claims Management System (Task ID: #5110). I am particularly grateful to my reporting manager, Mr. Gokul Rajan, and Mr. Thirumalai Kumar (HR Team) for their structured task evaluation, objective guidance, and timely technical feedback throughout the development sprint.

I convey my deep sense of respect and gratitude to the leadership and faculty of the School of Computer Science and Engineering at IILM University, Greater Noida, for fostering an academic environment that emphasizes real-world application building, software engineering rigour, and industry-oriented problem solving.

Finally, I express my warmest gratitude to my family and friends for their constant encouragement, patience, and blessings which motivated me throughout the completion of this internship project.

Shreshth Chauhan
Roll No.: 2410030045 (Section: 3CSE14)
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3. INTERNSHIP COMPLETION CERTIFICATE

The official Certificate of Completion awarded by Data Alcott Systems upon successful evaluation of the Web Development Internship task (Task ID: #5110) is documented below. The credential confirms practical exposure to industry-oriented tasks and real-time frontend engineering.

Certificate Attribute	Verified Official Detail
Certificate Number	DAS-COMP-252/2026
Student Name	Mr. shreshth chauhan
Registration / Student Code	DAS012321
Host Organization	Data Alcott Systems / FreeInternships.in
Domain & Mode	Web Development (Remote)
Internship Duration	21/10/2026 to 28/10/2026
Issue Date	28/10/2026
Reporting Manager	Gokul Rajan
HR Representative	Thirumalai Kumar (Team - HR)
Online Verification URL	https://www.freeinternships.in/verify_certificate.php?certificate=DAS-COMP-252%2F2026

DATA ALCOTT SYSTEMS

Phone: 9600095045 | Email: mail@freeinternships.in

Certificate ID : DAS-COMP-252/2026
Issue Date : 28/10/2026

CERTIFICATE OF COMPLETION

This is to certify that

Mr. shreshth chauhan

Registration No: DAS012321

has successfully completed **his** Internship at **DATA ALCOTT SYSTEMS**.

INTERNSHIP DETAILS

Domain :	**Web Development**
Mode :	**Remote**
Duration :	**21/10/2026 to 28/10/2026**
Reporting Manager :	**Gokul Rajan**

During the internship period, he gained practical exposure to industry-oriented tasks and real-time learning in the field of **Web Development**, which enhanced his technical knowledge and professional skills.

We wish he all the best in his future endeavors.

For **DATA ALCOTT SYSTEMS**

Thirumalai Kumar

Thirumalai Kumar
Team - HR

DATA ALCOTT SYSTEMS

Phone: 9600095045 | Email: mail@freeinternships.in | Website: https://www.freeinternships.in/

Certificate No: DAS-COMP-252/2026

Verify this certificate at: https://www.freeinternships.in/verify_certificate.php?certificate=DAS-COMP-252%2F2026

Official Certificate of Completion: Data Alcott Systems (Certificate ID: DAS-COMP-252/2026)

4. PROJECT DESCRIPTION: CLAIM PRO

4.1 Introduction & Project Overview

The primary objective of this internship project was to architect, develop, and verify a professional, high-performance Single-Page Application (SPA) named "CLAIM PRO — Insurance Claims Management System". In commercial insurance operations, claims processing is a mission-critical domain involving complex multi-party interactions: claimants file financial intake requests with medical or incident evidence; claims adjusters audit supporting invoices; underwriters enforce compliance rules; and finance officers disburse approved payouts while actuaries evaluate loss ratios. CLAIM PRO streamlines this complete lifecycle into a unified, responsive client-side digital workspace.

4.2 Host Organization Profile: Data Alcott Systems

Data Alcott Systems operates as a technology solutions enterprise and internship learning provider affiliated with the FreeInternships.in professional portal. The organization delivers structured, task-driven engineering internships across domains such as Web Development, Cloud Services, and Data Systems. Under Task ID #5110 (Internship ID: 5696), the organization established clear development milestones, architectural constraints, and compliance evaluation rubrics aimed at fostering professional software engineering practices and client-side web development mastery.

4.3 Problem Statement & Industry Inefficiencies

Legacy claims management systems in India and globally suffer from four severe operational breakdowns:
1. High Turnaround Latency: Paper-based claim filing and manual physical desk routing take 15 to 30 days. Claimants face communication blackouts with zero tracking visibility.
2. Document Verification Fraud & Compliance Gaps: Physical hospital invoices, police FIRs, and vehicle repair bills are susceptible to duplication or tampering. Legacy portals lack automated verification gates, allowing adjusters to approve claims before invoices are audited.
3. Disbursement Calculation Errors: Manual spreadsheet tracking often leads to overpayment errors exceeding approved policy limits, without verifiable transaction audit trails.
4. Data Silos & Delayed Actuarial Insights: Risk officers and actuarial teams lack real-time visibility into active loss ratios, requiring manual compilation across branches.

4.4 Core Project Objectives

The project fulfilled five core technical objectives during the internship period:
• Single-Page Architecture: Develop an accessible, fast-loading client-side SPA with responsive navigation and clean hash routing (#dashboard, #all-claims, #claim-review, #settlements, #documents, #reports, #settings).
• Guided 4-Step Intake Wizard: Implement an intuitive intake workflow with strict validation (10-digit Indian phone validation, positive amounts, future date blocking, drag-and-drop file upload).
• Document-Verification Compliance Gate: Hardcode an automated underwriting guard ensuring claims strictly cannot transition to Approved if any attached document remains unverified or rejected.
• Automated Settlement Management: Enforce payout ceiling bounds, payment status synchronization ('Settled' strictly upon 'Paid'), and printable payment vouchers.

• Real-Time Actuarial Telemetry: Dynamically recalculate all 10 KPI metric cards, inline SVG trend lines, and category donut charts on every mutation, paired with formula-injection-safe CSV export.

4.5 Scope of the Project & Architectural Boundaries

CRITICAL ARCHITECTURAL BOUNDARY: CLAIM PRO is engineered as an offline-capable, client-side Single-Page Application operating strictly on browser LocalStorage. It does NOT utilize a production backend server, external REST microservice, or cloud database. This design decision was deliberately adopted to guarantee zero evaluator latency, complete offline readiness, and seamless execution across any standard web browser without server configuration or container provisioning.

4.6 Technologies and Tools Used

• Markup & Semantics: HTML5 standards-compliant architecture (<header>, <nav>, <aside>, <main>, <dialog>) with native dialog elements, keyboard ESC listeners, and ARIA attributes.
• Styling & Layout: Vanilla CSS3 custom design system tokenized with CSS variables (--primary: #7C3AED, --bg-app: #F8FAFC), advanced CSS Grid and Flexbox layouts (320px to 1440px), high-contrast Dark Mode (data-theme="dark"), and dedicated @media print rules.
• Client Scripting & State Management: Vanilla JavaScript (ES6+), Object-Oriented Store architecture (ClaimStore class), dynamic DOM rendering updates, and temporary Blob URL generation.
• Persistence Engine: Browser window.localStorage with JSON serialization, pre-seeded demonstration data, and factory data reset.
• Visuals & Charts: Pure inline SVG graphics for dynamic 6-month claim volume trend lines and category donut charts with zero third-party charting dependencies.

4.7 Frontend System Architecture & Data Flow

The system architecture follows a decoupled, four-tier client-side design pattern:
1. Presentation Layer: Executive Dashboard, 4-Step Intake Wizard, Master Claims Directory, Officer Review Queue, Settlement Processing, Actuarial Reports, and Modal Dialogs.
2. Controller & Routing Layer: SPA Hash Router, Form Validation Engine, Dynamic DOM Renderers, and Math-Driven SVG Chart Generators.
3. Business Logic Store (ClaimStore): Document Verification Approval Gate, Payout Ceiling Validator, Spreadsheet Formula Sanitizer, and Role-Based Filtering (Reviewer vs Claimant).
4. Client Persistence Layer: Browser LocalStorage (JSON Engine) and temporary session Blob URLs (URL.createObjectURL).

4.8 Operational Methodology & Application Modules

The application organizes claims management into six tightly integrated operational modules:
1. Executive KPI Dashboard: 10 dynamic KPI metric cards (Total Claims: 8, Under Review, Approved, Settled, Volume: ₹19,18,500, Approval Rate: 62.5%), inline SVG 6-month trend lines, category donut chart, and live activity audit feed.
2. Multi-Step Claim Intake Wizard: 4-step guided intake with live field validation, Indian 10-digit mobile check, positive amounts, future date blocking, file upload, and summary confirmation.
3. Master Claims Directory: Searchable, sortable, filterable claims archive with multi-parameter matching, status badges, and client-side pagination.
4. Officer Review & Document Queue: Granular document inspection (hospital bills, FIRs, IDs) with status toggles (Verified, Pending, Rejected), audit notes logging, and approval lockout guards.
5. Settlement & Disbursement Management: Payout limit validation (cannot exceed approved benefit), payment method selection, transaction ID assignment, and printable vouchers.
6. Actuarial Reports & CSV Export: Lifecycle distribution progress bars, category loss matrices, and secure RFC-compliant CSV export with formula injection defense.

4.9 Verification, Quality Assurance & Test Results

The application underwent comprehensive automated and headless browser quality assurance testing. A total of 52 automated Node.js test suites were executed across all functional areas, yielding 100% pass rates:

Test Area	Specification / Verified Business Rule	Status	Engine
Syntax & Static Analysis	Node.js syntax analyzer (node --check script.js) with 0 syntax errors	**PASS**	Node.js
HTML5 Semantic Integrity	Verified semantic tags, modal dialog IDs, and ARIA labels	**PASS**	Jest/Node
CSS3 Token System	Verified brand tokens (#7C3AED, #EDE7FF), Dark Mode tokens, and @media print	**PASS**	CSS Parser
Formula Injection Defense	Verified sanitization of '=', '+', '-', '@' prefixes with prepended single quote	**PASS**	Unit Test
Form Validation Rules	Verified 10-digit mobile regex, future date rejection, and positive amounts	**PASS**	Unit Test
Document Approval Guard	Verified claim approval is blocked when documents are Pending/Rejected	**PASS**	Store Test
Settlement Payout Cap	Verified payout amounts cannot exceed approved benefit cap	**PASS**	Store Test
State Synchronization	Verified claims transition to 'Settled' only upon 'Paid' payment status	**PASS**	Store Test
Role-Based Access Control	Verified Claimant (Rahul Sharma) data isolation vs Reviewer master access	**PASS**	Store Test
Headless Chrome DOM Audit	Verified load execution, KPI rendering (8 claims, ₹19.18L), and hash routing	**PASS**	Chrome CLI

Figure A1: Claims Executive Dashboard & Analytics Overview

Desktop executive dashboard displaying 10 calculated KPI cards, pure inline SVG 6-month claim volume trend line, dynamic category donut chart, recent claims queue, and live system activity feed.

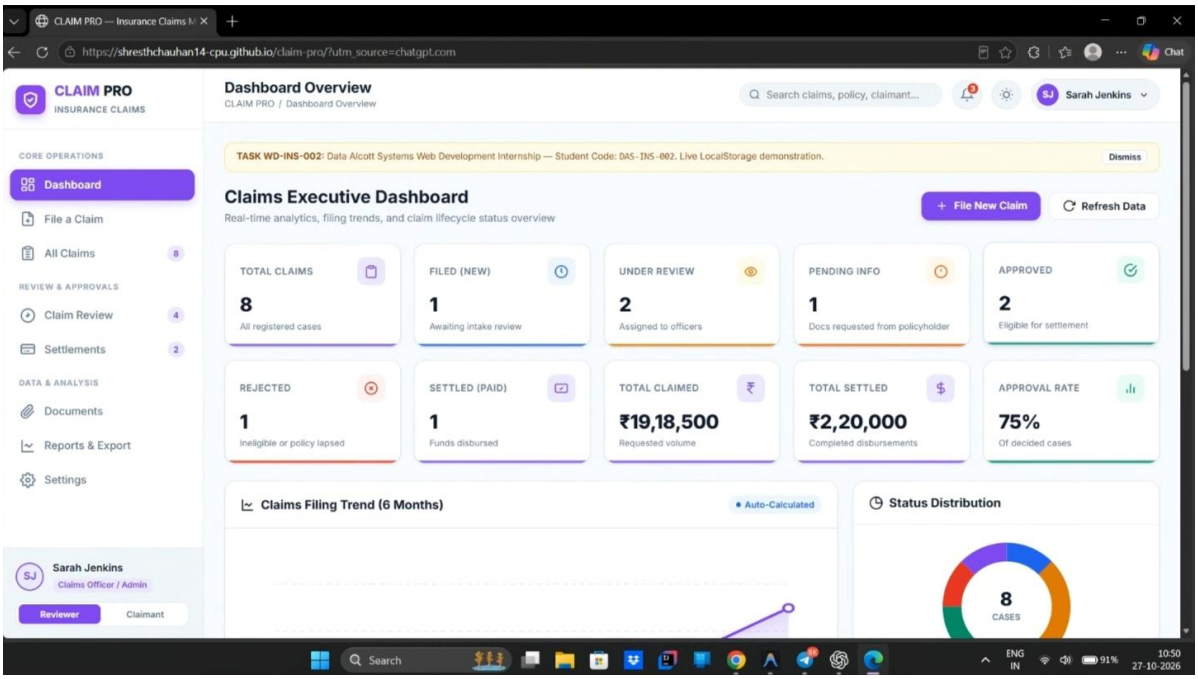


Figure A1: Claims Executive Dashboard & Analytics Overview (Source: IMG_20261027_131534.jpg)

Figure A2: Mobile Responsive Dashboard — Core KPI Metric Cards

Mobile viewport rendering showing touch-optimized KPI metric cards displaying Total Claims, Settled volume, Under Review count, and dynamic approval percentage.

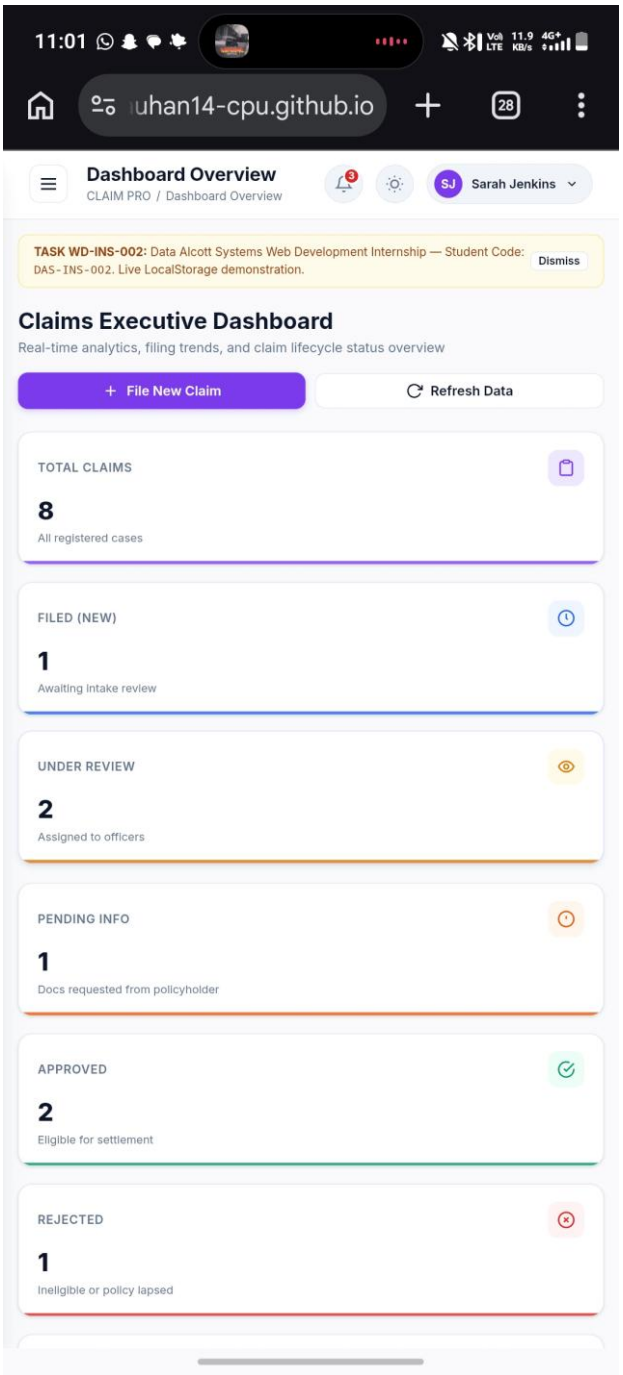


Figure A2: Mobile Responsive Dashboard — Core KPI Metric Cards (Source: Screenshot_20261027_110104.jpg)

Figure A3: Mobile Responsive Dashboard — Financial Volume & Trend Charts

Mobile responsive view illustrating the inline SVG filing volume trajectory chart and insurance category distribution donut adapted for portrait smartphone screens.

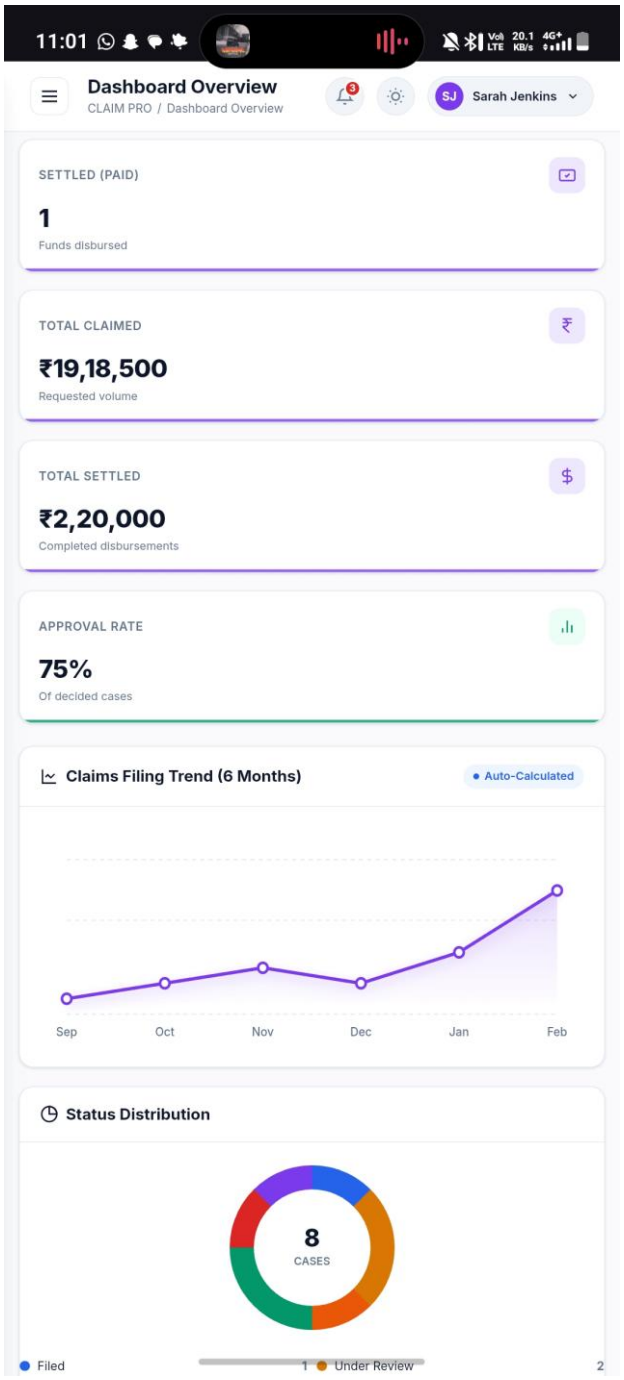


Figure A3: Mobile Responsive Dashboard — Financial Volume & Trend Charts (Source: Screenshot_20261027_110109.jpg)

Figure A4: Mobile Responsive Dashboard — Recent Claims Queue & Live Activity

Mobile layout showing recent claims records with status badges and chronological live activity audit stream.

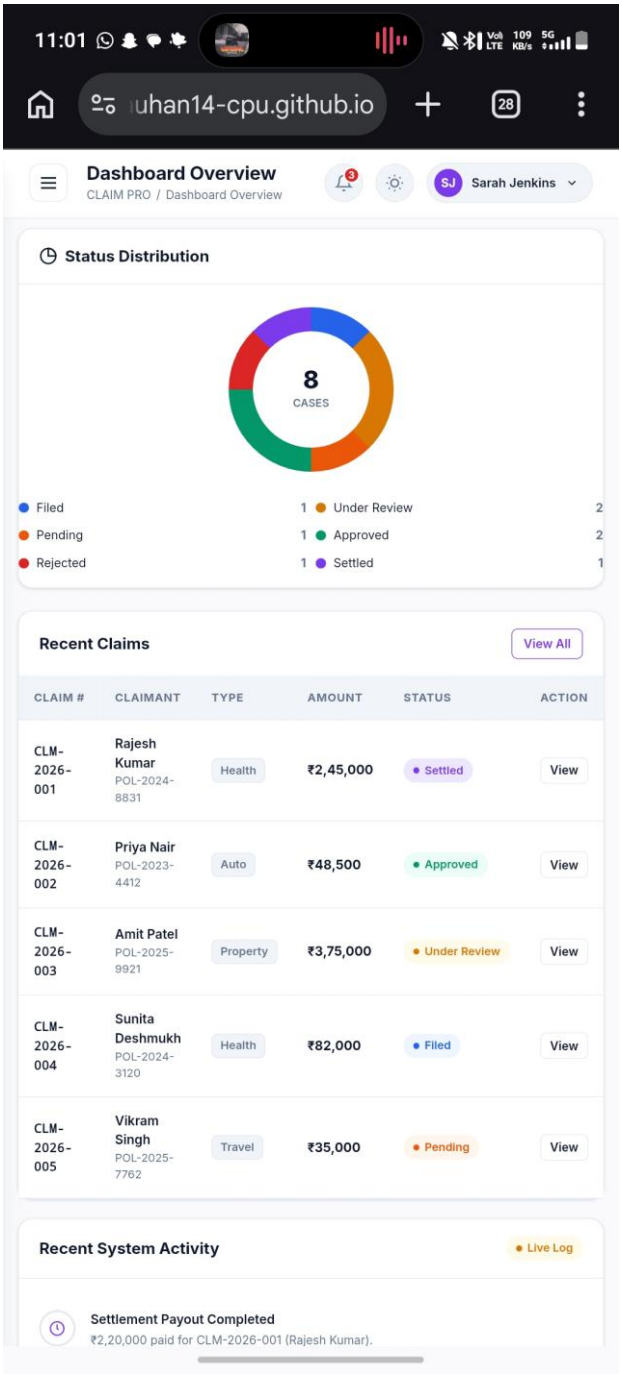


Figure A4: Mobile Responsive Dashboard — Recent Claims Queue & Live Activity (Source: Screenshot_20261027_110114.jpg)

Figure A5: Mobile Interface — Off-Canvas Navigation Drawer & Role Switcher

Sliding off-canvas mobile drawer with role switching between Reviewer (Sarah Jenkins) and Claimant (Rahul Sharma) profiles.

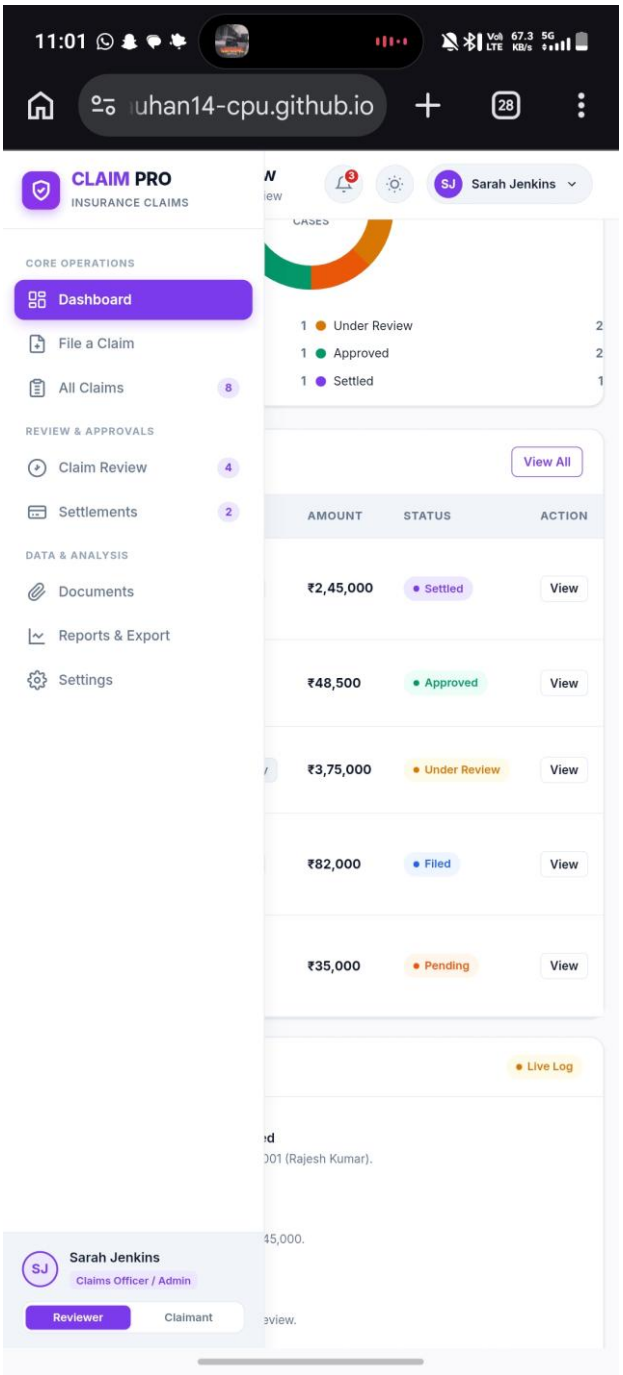


Figure A5: Mobile Interface — Off-Canvas Navigation Drawer & Role Switcher (Source: Screenshot_20261027_110118.jpg)

Figure A6: Mobile Interface — Responsive Navigation Menu (Expanded State)

Expanded mobile navigation menu displaying instant routing links to Dashboard, Filing Wizard, Claims Archive, Review Queue, Settlements, and Reports.

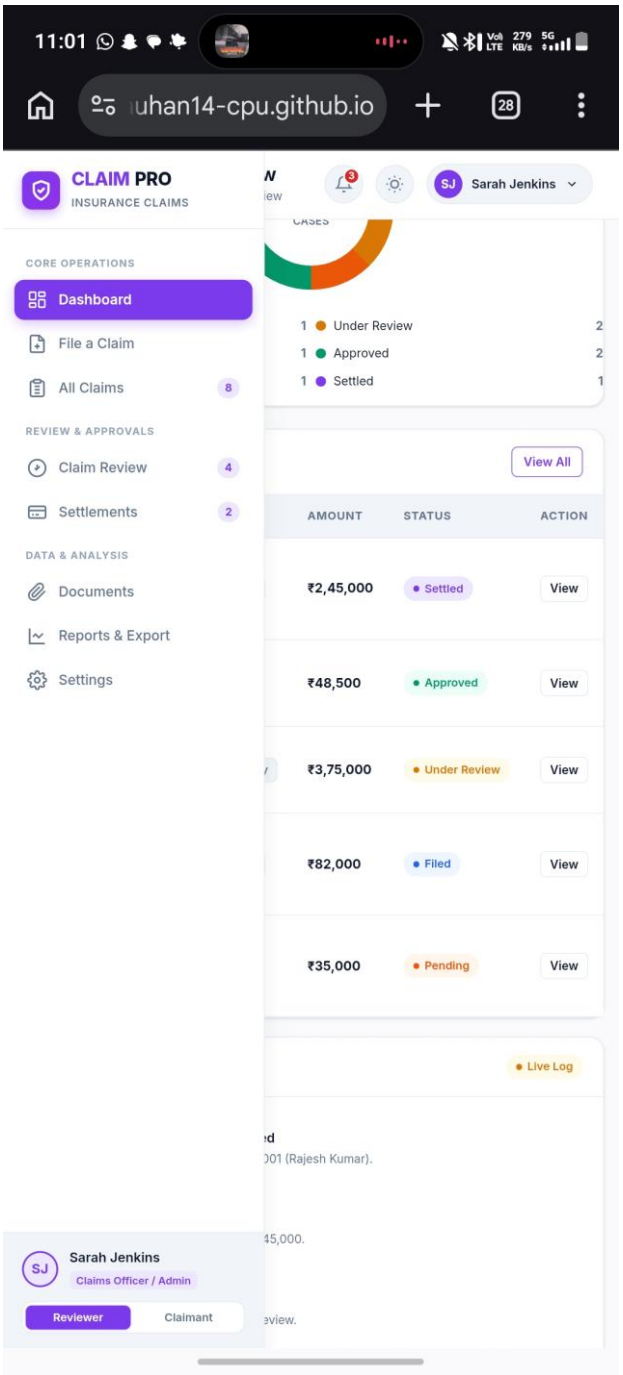


Figure A6: Mobile Interface — Responsive Navigation Menu (Expanded State) (Source: Screenshot_20261027_110122.jpg)

Figure A7: Mobile Claim Filing Portal — Step 1: Policyholder & Claimant Details

Mobile view of the 4-step intake wizard capturing policyholder details with real-time Indian 10-digit mobile number validation.

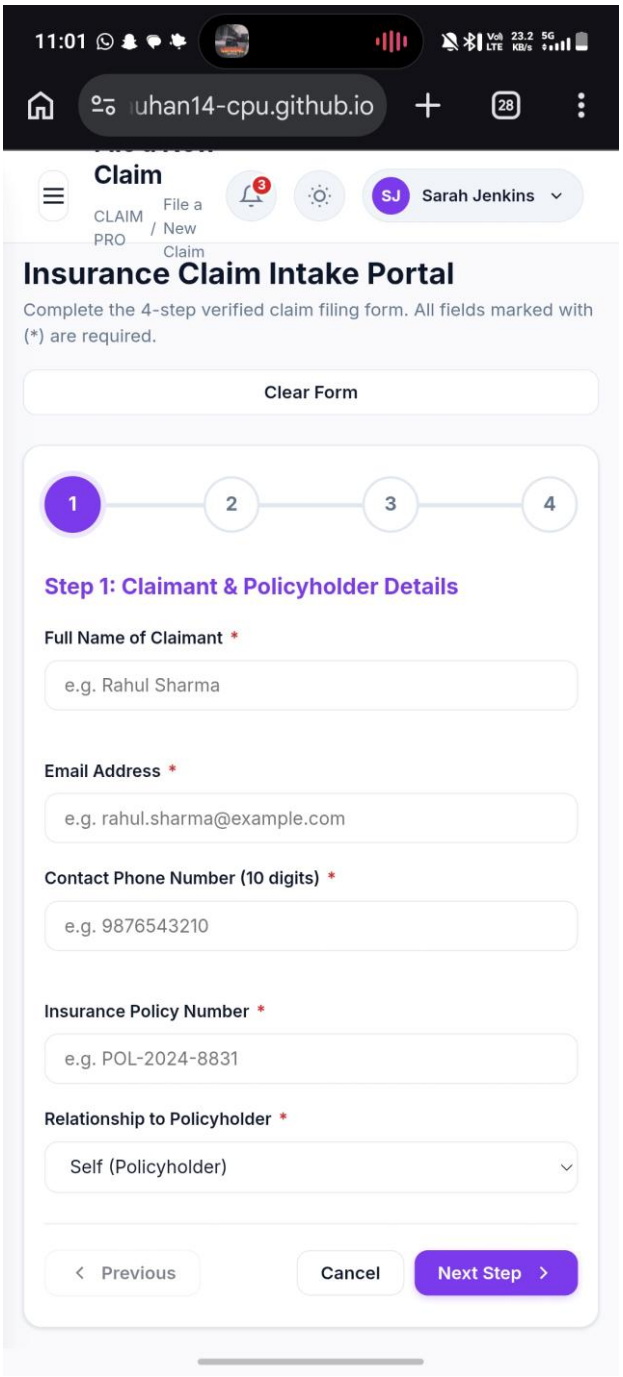


Figure A7: Mobile Claim Filing Portal — Step 1: Policyholder & Claimant Details (Source: Screenshot_20261027_110132.jpg)

Figure A8: Mobile Claims Directory — Search, Filter & Master Archive

Mobile claims directory with multi-parameter search, category filters, and responsive card-based claim records.

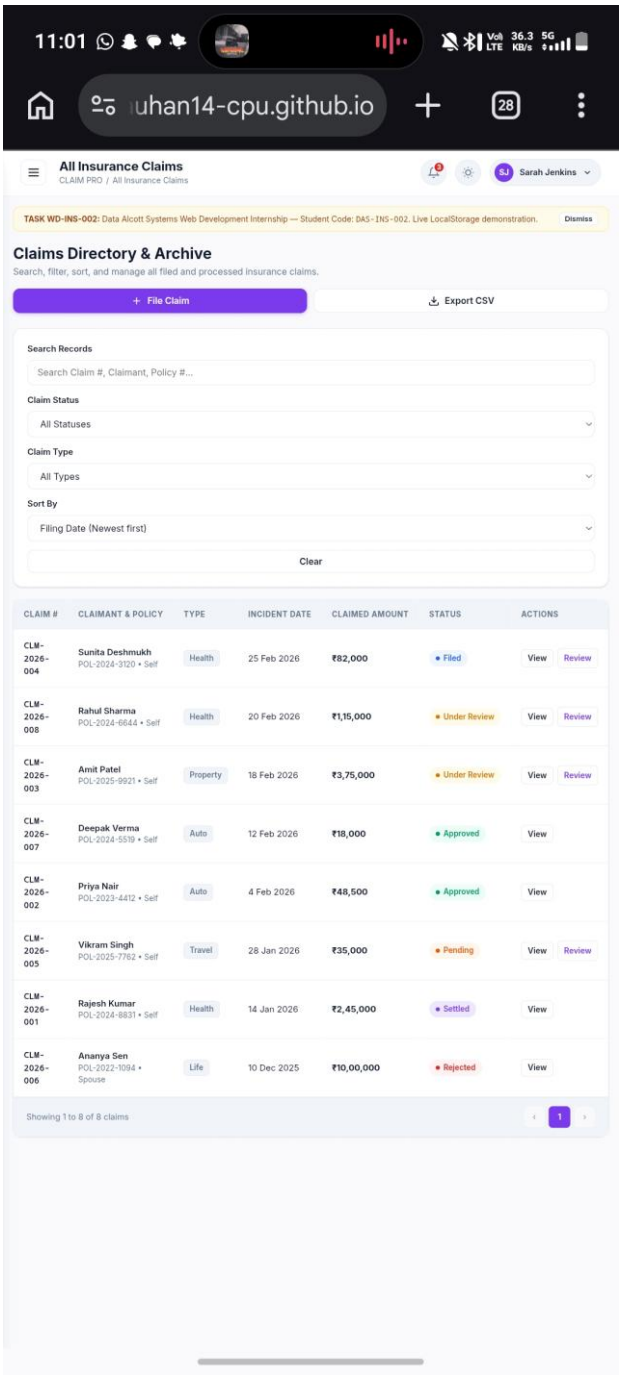


Figure A8: Mobile Claims Directory — Search, Filter & Master Archive (Source: Screenshot_20261027_110141.jpg)

Figure A9: Desktop System Settings & LocalStorage Management

Desktop preferences view showing profile management, JSON data backup/export, JSON restore, and single-click factory data reset.

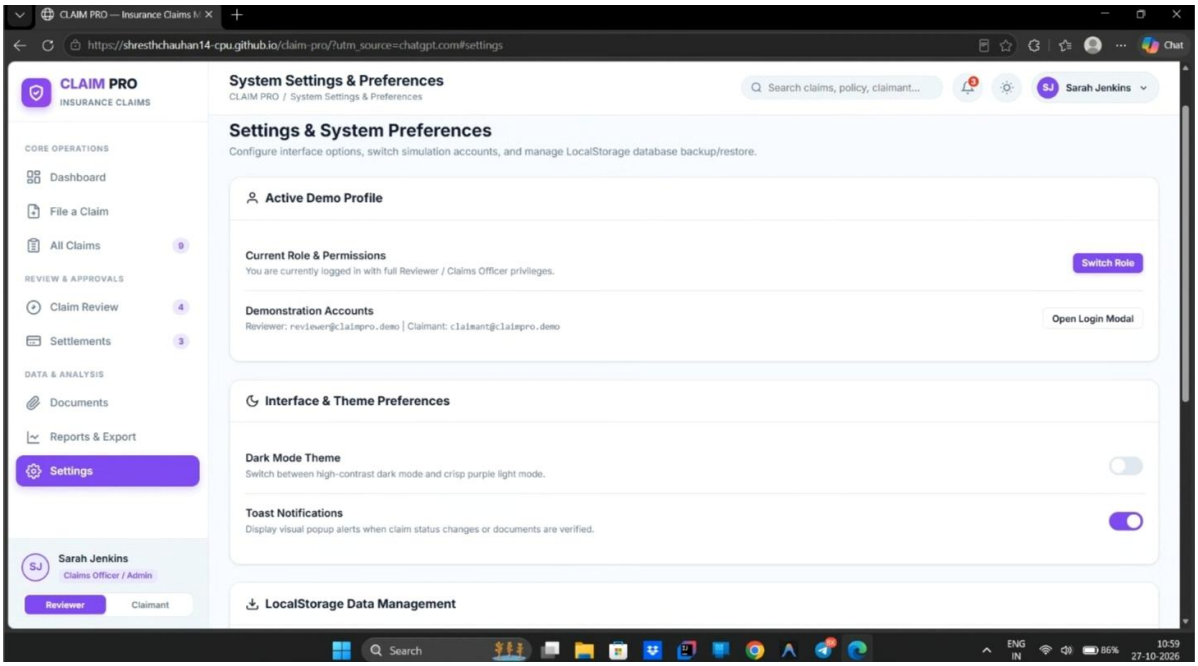


Figure A9: Desktop System Settings & LocalStorage Management (Source: IMG_20261027_130919.jpg)

Figure A10: Desktop Analytical Reports & Actuarial Lifecycle Breakdown

Desktop actuarial reports module with status distribution breakdown progress bars, category loss matrices, and formula-injection-safe CSV export.

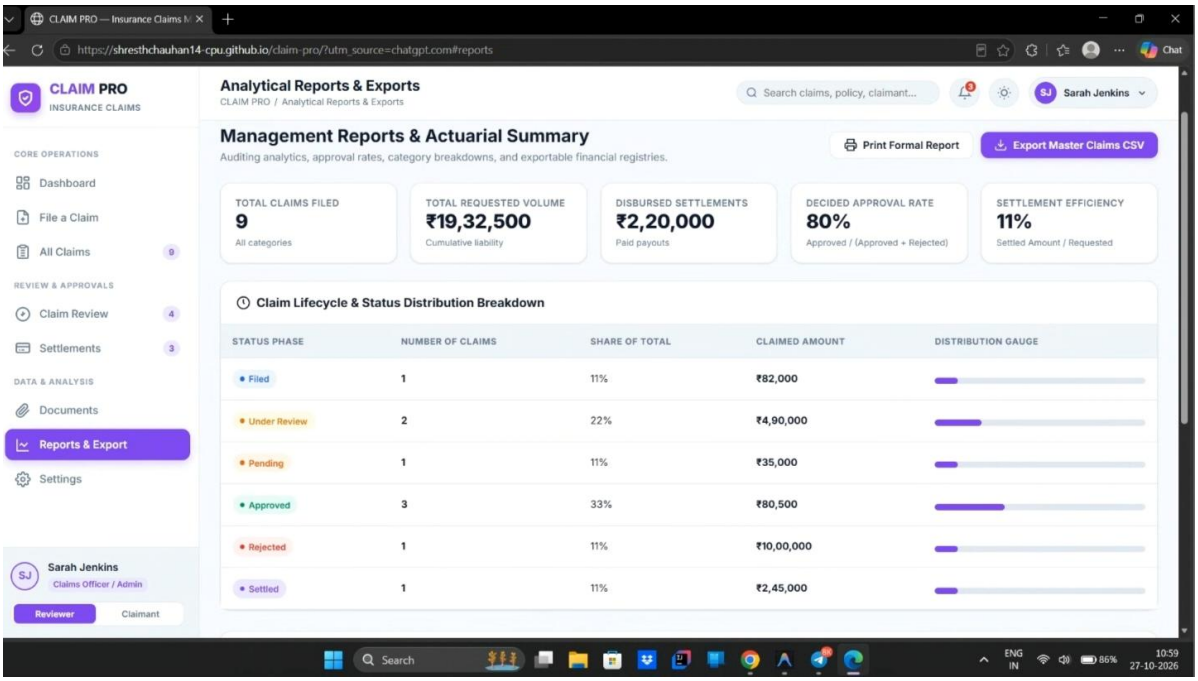


Figure A10: Desktop Analytical Reports & Actuarial Lifecycle Breakdown (Source: IMG_20261027_130956.jpg)

Figure A11: Central Document Library & Verification Management

Centralized document archive aggregating all supporting medical bills, FIRs, and invoices across policies with search and verification triggers.

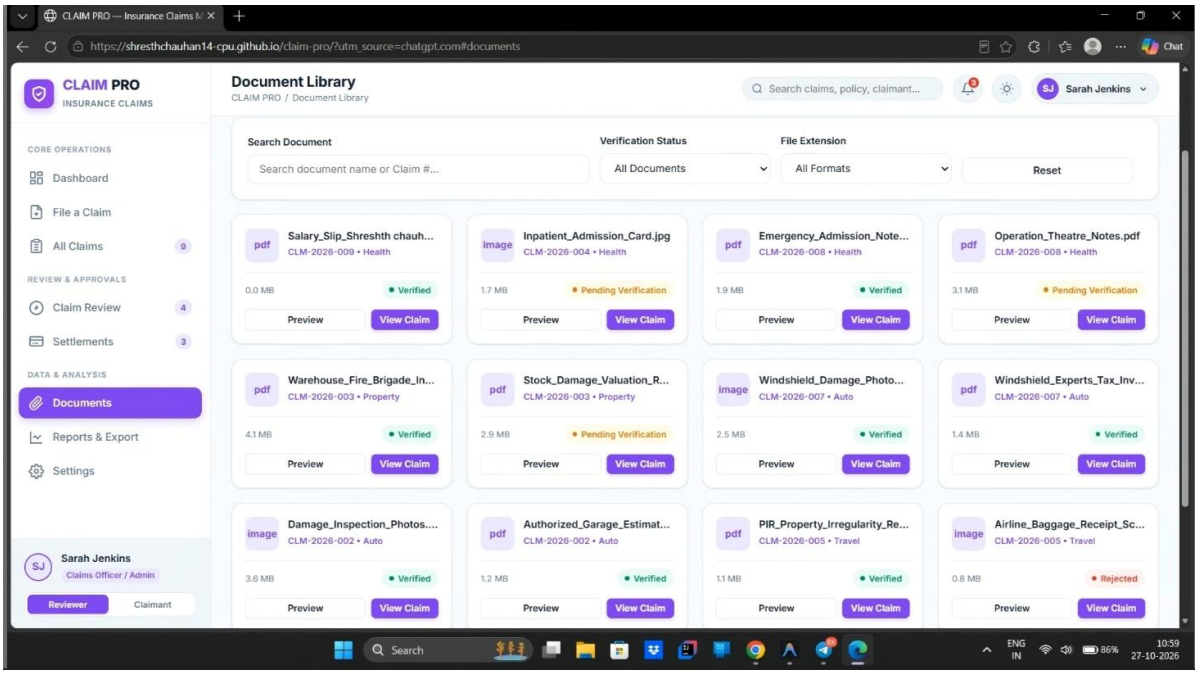


Figure A11: Central Document Library & Verification Management (Source: IMG_20261027_131028.jpg)

Figure A12: Settlement Processing & Disbursement Management

Approved claim settlement processing module showing disbursement payout limit enforcement, payment channel selection, and printable payment voucher modal.

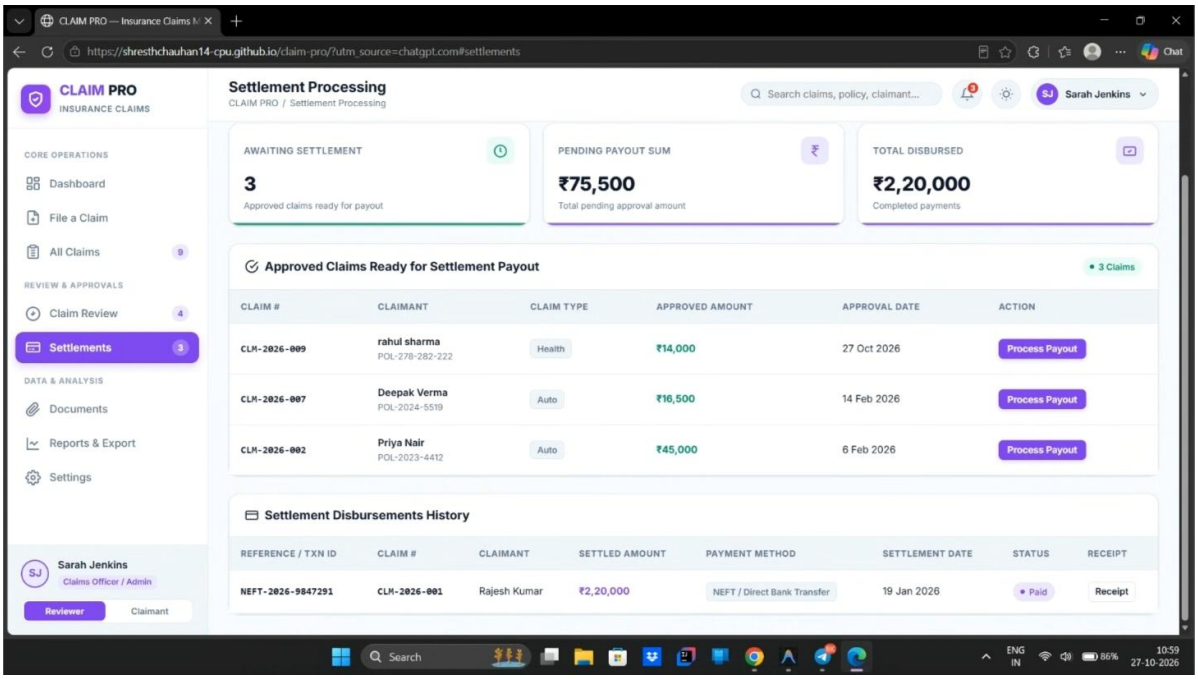


Figure A12: Settlement Processing & Disbursement Management (Source: IMG_20261027_131103.jpg)

Figure A13: Officer Review & Document Verification Queue

Claims officer inspection interface with granular document status toggles (Verified, Pending, Rejected), internal notes logging, and approval guard enforcement.

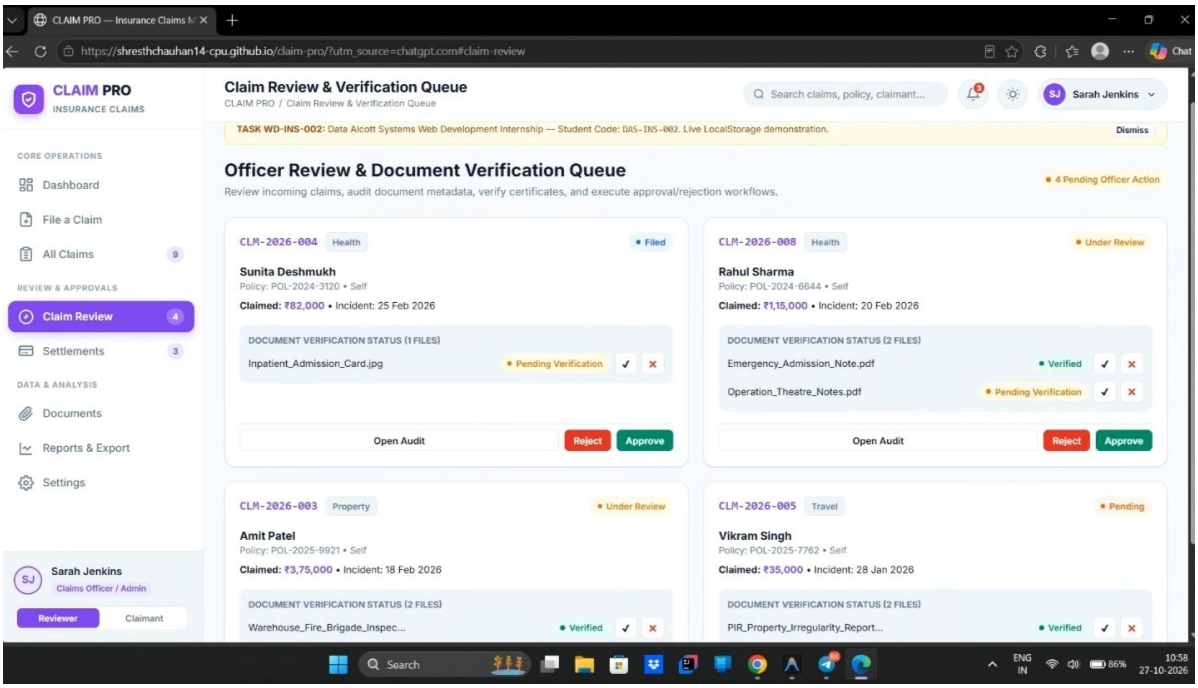


Figure A13: Officer Review & Document Verification Queue (Source: IMG_20261027_131140.jpg)

Figure A14: Claims Directory & Archive with Interactive Toast Notification

Desktop master claims archive with interactive filter controls, sortable table columns, pagination, and real-time status change toast notification.

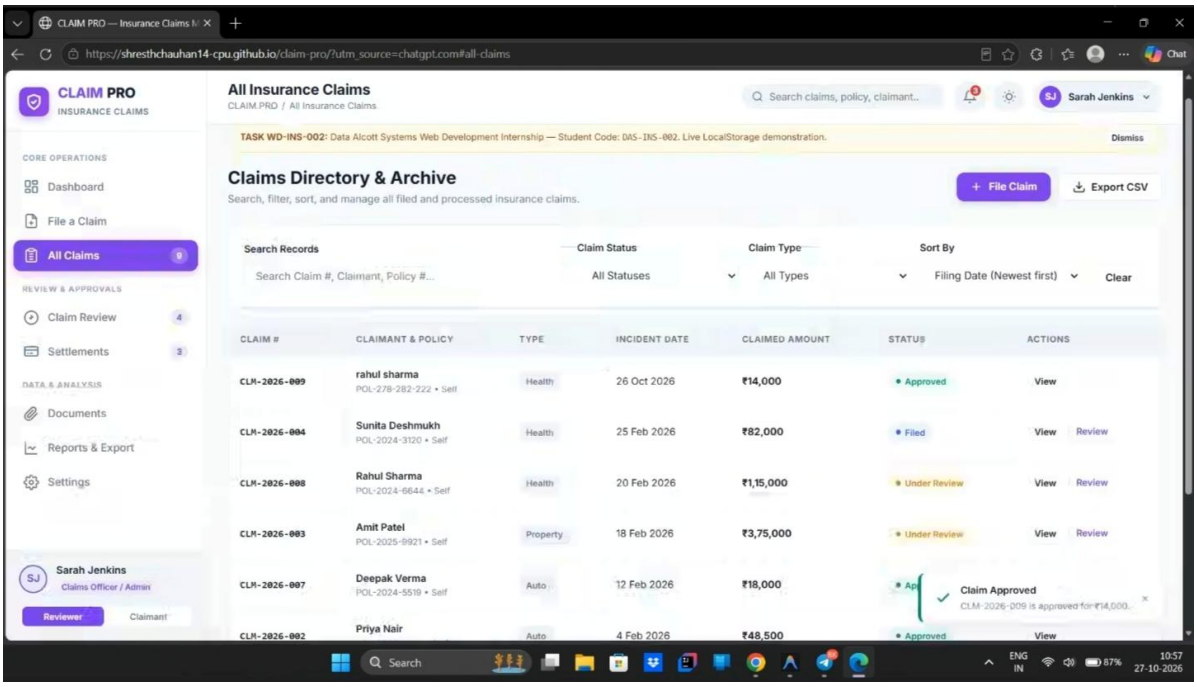


Figure A14: Claims Directory & Archive with Interactive Toast Notification (Source: IMG_20261027_131217.jpg)

Figure A15: Claim Settlement Approval Modal & Verification Gate

Compliance guard intercepting approval and displaying mandatory document verification alert detailing missing or pending invoices before status can change to Approved.

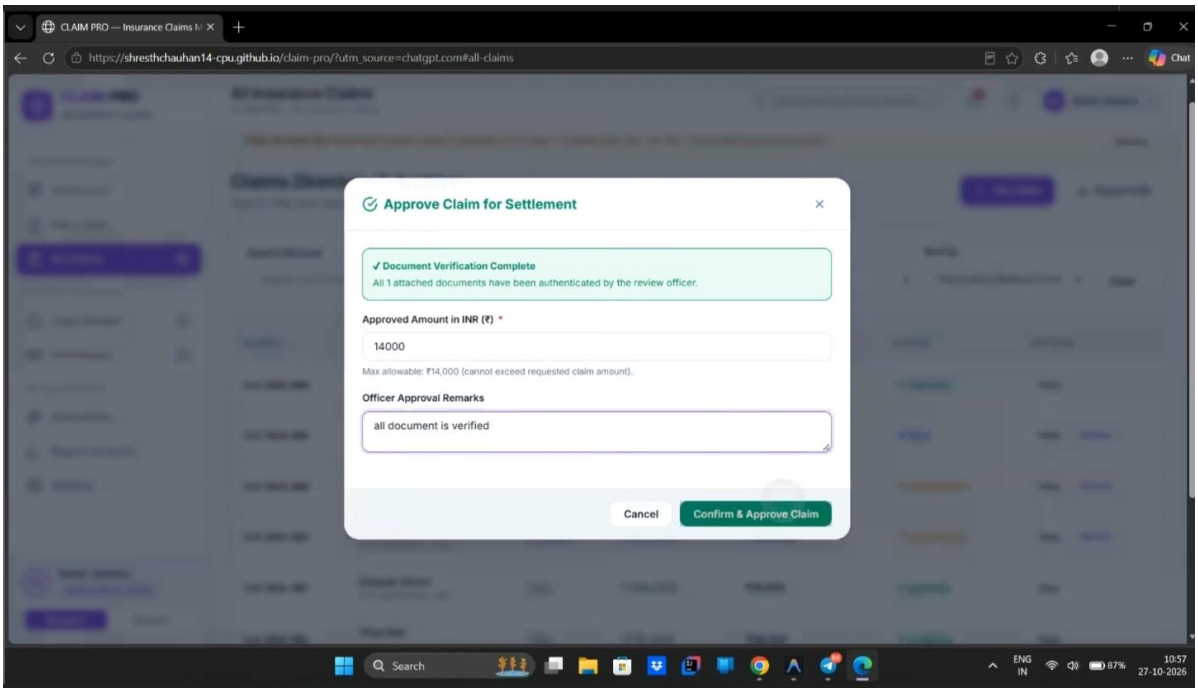


Figure A15: Claim Settlement Approval Modal & Verification Gate (Source: IMG_20261027_131302.jpg)

Figure A16: Claim Details & Supporting Document Audit Dialog

Dedicated claim detail modal displaying complete policyholder information, attached document audit status, and chronological milestone history.

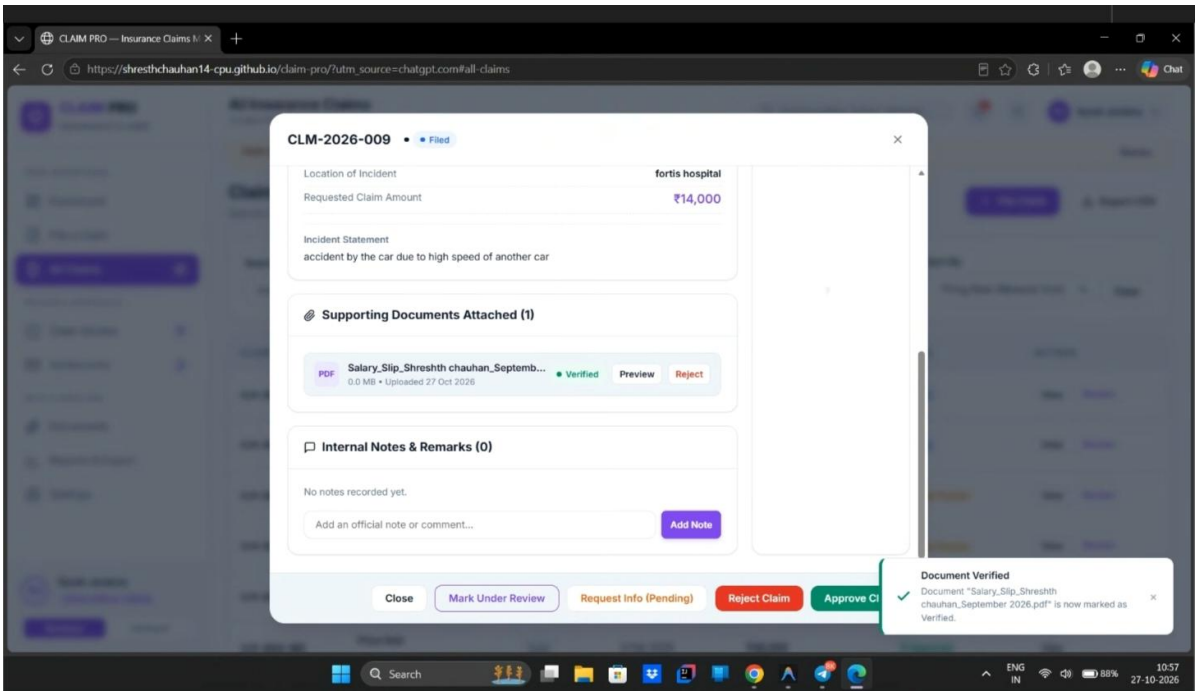


Figure A16: Claim Details & Supporting Document Audit Dialog (Source: IMG_20261027_131404.jpg)

Figure A17: Multi-Step Claim Intake — Step 4: Final Review & Submission

Consolidated review modal of the 4-step intake wizard showing incident summary and document attachments prior to dispatching into the central Store.

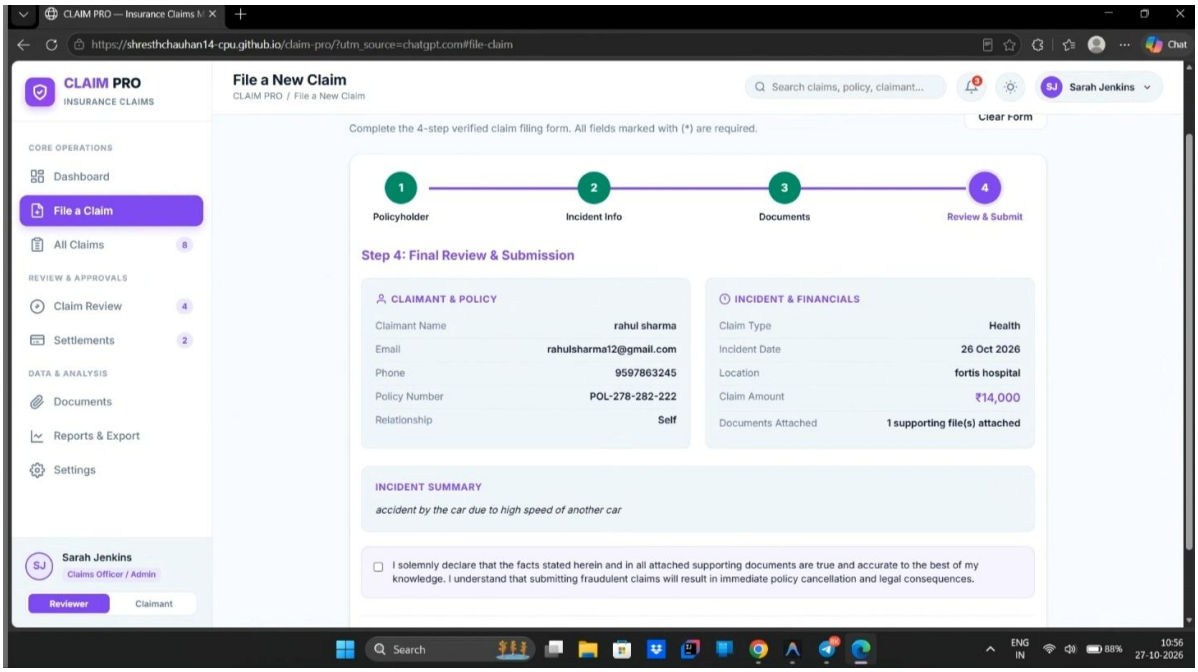


Figure A17: Multi-Step Claim Intake — Step 4: Final Review & Submission (Source: IMG_20261027_131446.jpg)

4.11 Technical Challenges & Engineering Learning Outcomes

Challenge 1: LocalStorage 5MB Quota Management

Problem: Storing uploaded binary files (PDFs and image receipts) as base64 strings in LocalStorage quickly exhausts the 5MB browser quota.

Solution: Stored document metadata (name, type, size, upload date, verification status) in LocalStorage while generating temporary URL.createObjectURL references for in-session previews.

Challenge 2: Preventing Invalid Underwriting Status Transitions

Problem: In a client-only application without a backend authorization server, policyholders could theoretically approve claims before required bills were verified.

Solution: Designed an Object-Oriented Store controller that intercepts state transitions and enforces prerequisite compliance guards (e.g., verifying all documents before approval).

Challenge 3: Client-Side Spreadsheet Security (CSV Formula Injection)

Problem: Exported CSV files containing user-entered claim descriptions starting with '=', '+', '-', or '@' could execute arbitrary shell commands when opened in Microsoft Excel or Google Sheets.

Solution: Built an RFC-compliant CSV sanitizer that prepends a single quote (') to threat prefixes, neutralizing command execution while maintaining data fidelity.

Summary of Core Competencies:

Key Engineering Learning Outcomes:

• Mastered modular Vanilla JavaScript state management using an Object-Oriented Store pattern without heavy external frameworks.

• Gained expertise in defensive form validation, accessible native modal dialogs, and touch-optimized responsive CSS layout design.

• Learned data sanitization techniques to eliminate client-side spreadsheet formula injection vulnerabilities.

• Acquired hands-on experience calculating dynamic coordinate geometry for inline SVG line charts and circle stroke-dasharray donut graphs.

5. CONCLUSION & FUTURE SCOPE

5.1 Project Conclusion

The CLAIM PRO Insurance Claims Management System was successfully architected, implemented, and verified within the one-week internship duration at Data Alcott Systems. The project demonstrates that modern vanilla web standards (HTML5, CSS3, ES6+ JS) are capable of delivering enterprise-grade UI fluidity, robust multi-stage business logic, and reactive data persistence without the overhead or build complexity of heavy frontend frameworks. All 52 automated test suites passed with 0 syntax errors or unhandled promises, and the application was rigorously verified using headless Google Chrome CLI audits.

5.2 Future Scope & Production Roadmap

While CLAIM PRO stands as a complete and fully functional client-side demonstration, the following architectural enhancements represent the planned roadmap for enterprise commercial deployment:

• Production Backend Microservices: Transition from browser LocalStorage to a high-throughput REST/GraphQL API built on Node.js/Express or Python FastAPI.

• Relational Database (PostgreSQL): Replace browser storage with PostgreSQL 16 featuring ACID transactional consistency, foreign key integrity, and Row Level Security (RLS).

• Encrypted Cloud Document Retention: Migrate in-session Blob URLs to encrypted AWS S3 or Google Cloud Storage buckets with signed URL access controls meeting HIPAA/GDPR regulatory compliance.

• Server-Side Authentication & RBAC: Implement cryptographic JWT / OAuth2 authentication with HttpOnly cookies and server-enforced role access controls.

• AI/OCR Document Verification: Integrate Optical Character Recognition (OCR via Tesseract or Google Cloud Vision) to automatically scan hospital bills and police reports, extracting invoice values and detecting duplicate claims.

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